

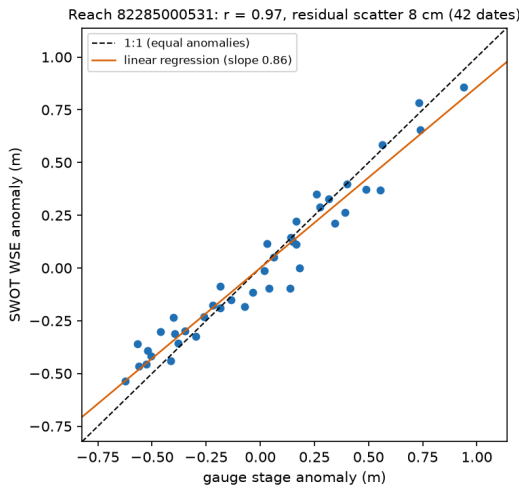
Hydropower screening from SWOT satellite observations on the Athabasca River

Jamie Williams · Mechanical Engineering, Carleton University · September 2026 · Code and data: github.com/jamiewilliams07/river-power

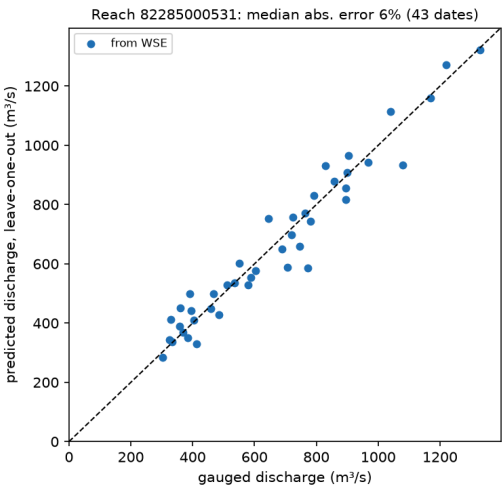
The gross hydropower potential of a river reach is $P = \rho g Q H \eta$, which requires discharge Q and hydraulic head H . I tested whether SWOT satellite observations (NASA/CNES, with the Canadian Space Agency) can provide both along the Athabasca River at Fort McMurray, using Water Survey of Canada gauge 07DA001 as the reference. **A rating curve fitted to SWOT water-surface elevation predicted gauged discharge with a median error of 5-6% under leave-one-out cross-validation at two independent reaches.** SWOT width showed no usable relationship with discharge. Using SWOT for both discharge and head, the design flow is within 2% of the gauged value at the gauge reach and a median 3% across 8 reaches upstream. Accuracy decreases near the Clearwater confluence: 22% and 12% median error at the two nearest reaches, compared with 7–9% further upstream. SWOT slope places about 106 m of head in the 9 reaches of rapids above Fort McMurray. The screening shortlists 3 reaches, led by one 2923 km from the river mouth (87 MW, 369 GWh/yr gross theoretical potential).

1. Does SWOT measure the river accurately?

	Gauge reach	Reach 40 km downstream
SWOT observations matched to gauge days	43	36
WSE vs gauge stage anomaly: correlation / scatter	0.97 / 8 cm	0.99 / 11 cm
Discharge error, leave-one-out (median)	6.1%	4.8%
Discharge error, latest 30% of dates (median)	5.2%	5.1%
Width vs discharge: correlation	0.03	0.21

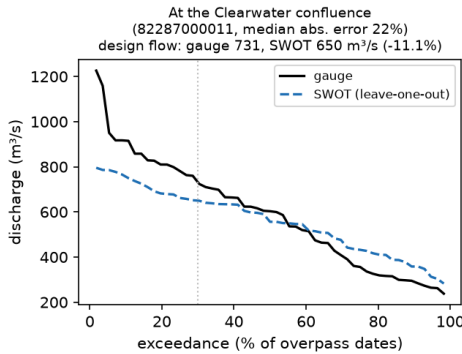
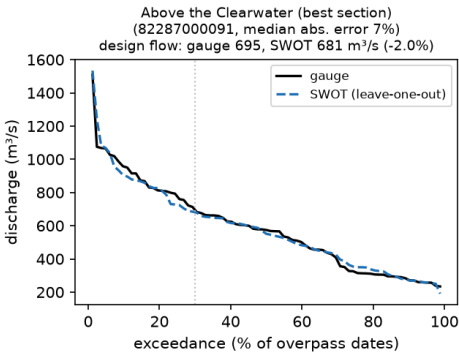
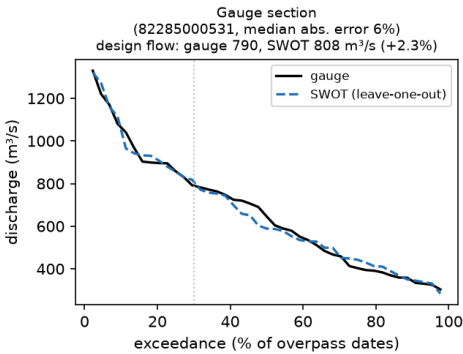


(a) SWOT WSE anomaly vs gauge stage anomaly at the gauge reach.



(b) Discharge predicted from SWOT WSE (leave-one-out) vs gauged discharge.

Flow-duration curves: SWOT vs gauge (coincident overpass dates)

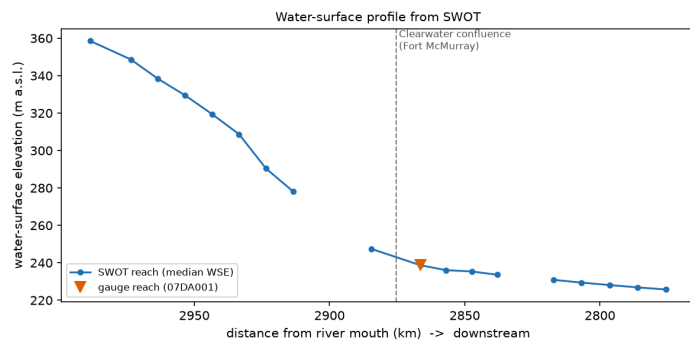


(c) Flow-duration curves from SWOT and the gauge on coincident dates at three reaches. The design flow is the discharge exceeded 30% of the time.

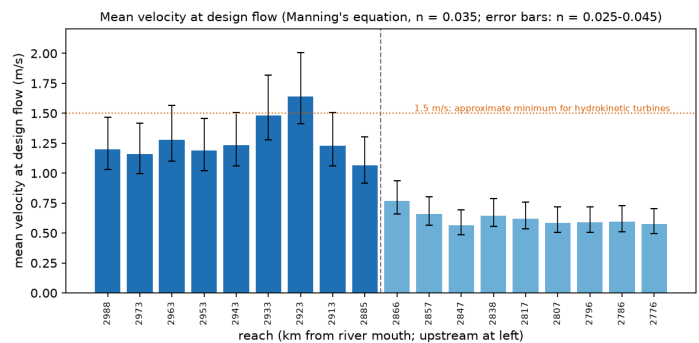
Method: measurement

- Data.** SWOT River Single Pass observations (Jul 2023–Sep 2026, versions C and D) from NASA's Hydrocron API for 19 reaches (18 with sufficient valid observations); daily discharge and stage from Water Survey of Canada gauges 07DA001 and 07CD001 (record from 1957).
- Quality control.** 6,612 observations, 1,299 retained. Removed ice-season, flagged, partial, dark-water and duplicate observations, and outliers.
- Version consistency.** SWOT product version D (SWORD v17b) reuses some reach IDs for different locations, giving WSE differences of 13–20 m for the same pass. After matching reaches by coordinates, all 19 agree within 9 cm.
- Rating curve.** $Q = a(h - h_0)^b$ fitted to SWOT WSE ($0.8 \leq b \leq 5$); evaluated by leave-one-out cross-validation and a chronological 70/30 split.
- Head.** SWOT slope $\times$ reach length, cross-checked against WSE differences between adjacent reaches: the two agree within 0.5%.

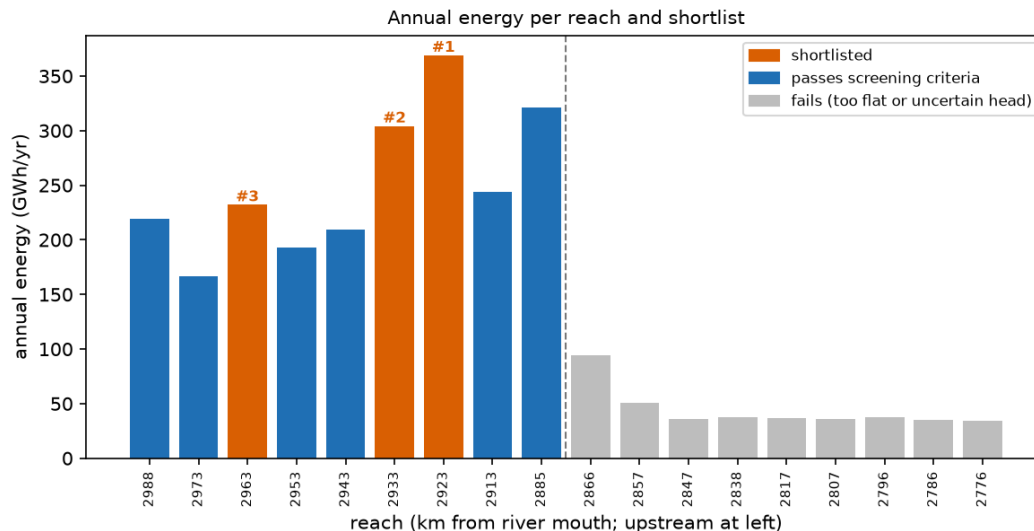
2. Where is the hydropower potential?



(d) Water-surface profile from SWOT. Gaps indicate missing reaches.



(e) Mean velocity at the design flow from SWOT width and slope (Manning).



Criteria: head gradient ≥ 0.5 m/km; ≥ 30 clean overpasses; SWOT head spread $\leq 30\%$. Ranked by annual energy per km.

(f) Annual energy per reach. Orange: shortlisted; blue: meets the criteria; grey: insufficient head gradient or measurement confidence.

Shortlist: reaches recommended for site assessment

Rank	Reach (km from mouth)	Head	Head gradient	Capacity	Annual energy	Capacity factor	Mean velocity
#1	82287000041 (2923 km)	17.4 m	1.74 m/km	87 MW	369 GWh/yr	0.48	1.6 m/s
#2	82287000051 (2933 km)	14.3 m	1.44 m/km	72 MW	304 GWh/yr	0.48	1.5 m/s
#3	82287000081 (2963 km)	10.9 m	1.10 m/km	55 MW	232 GWh/yr	0.48	1.3 m/s

Method: screening

- SWOT-only discharge.** Design flow, SWOT vs gauge (coincident dates): 808 vs 790 m^3/s (+2.3%) at the gauge reach; median error 8% across 9 upstream reaches (07DA001 minus 07CD001). Power has the same relative error.
- Power.** $P = \rho g Q H \eta$; Q = design flow, exceeded 30% of the time (790 m^3/s ; 639 m^3/s above the Clearwater); $\eta = 0.80$.
- Annual energy.** Daily discharge, 66 years; environmental flow 10% of mean; turbine off below 15% of design flow. Capacity factor = mean / installed output.
- Velocity.** Manning's equation (wide channel) with SWOT width and slope; roughness coefficient $n = 0.035$ (range 0.025–0.045).
- Criteria.** Head gradient ≥ 0.5 m/km; ≥ 30 valid observations; head spread (P10–P90)/median $\leq 30\%$. 9 of 18 reaches pass; ranked by annual energy per km.

Limitations

- Gross theoretical potential only: no costs, access, ice, fish passage or land use.
- SWOT-derived discharge is less accurate near the Clearwater confluence (22% and 12% error). Backwater from the Clearwater is a plausible cause, but a test for it was inconclusive.
- The rating curve requires calibration against a gauge.
- Velocity depends on an assumed roughness coefficient; no velocity data were used.
- SWOT observations cover the open-water season (May–Oct) only.
- Two reaches are missing from the profile, so total head is understated.

Conclusion. On this river, SWOT WSE is a far better predictor of discharge than channel width, and SWOT slope alone locates the head. Next steps: extend the validation to other gauged Canadian rivers, and test CSA RADARSAT Constellation imagery on rivers narrower than SWOT's ~100 m limit.